

2 Exponents and Powers

WARM UP

1. **Find the value of the following:**
   a. $(-1)^7$ b. 13^3 c. $\left(\frac{-5}{-3}\right)^5$ d. $\left(\frac{-7}{11}\right)^4$
2. **Express the following in the power notation.**
   a. –15,625 b. 2,197 c. 1,024 d. 4,096
3. **Simplify the following:**
   a. $\{(2 \times 5)^3 + (5 \times 3)^2\}^2$ b. $\{(3)^{-3} - 5^3\} \div 6^0$ c. $3^2 \times 2^{-3} \times \left(\frac{1}{4}\right)^{-2}$ d. $4 \times \left(\frac{2}{3}\right)^4 \times 9$
4. **Evalute the following using the laws of exponents.**
   a. $(5^3 \times 6^3)$ b. $\left(\frac{-2}{3}\right)^3 \times \left(\frac{-2}{3}\right)^4$ c. $\left(\frac{-3}{-7}\right)^2 \times \left(\frac{3}{7}\right)^3$ d. $(-5)^4 \times 3^5 \div 2^{-4}$
5. **Find the value of the following:**
   a. $\{(5)^3\}^2$ b. $\{(x)^a\}^{-a}$ c. $\{(a^3)^3\}^6$ d. $\left\{\left(\frac{-2}{-3}\right)^3\right\}^3$
6. **Simplify the following and express as a single power.**
   a. $3^2 \times 5^3$ b. $9^5 \times 6^5$ c. $a^4 \times b^4 \times c^4$ d. $7^3 \times 3^3$
7. **Simplify the following:**
   a. $(2 \times 9)^3$ b. $\left\{\left(\frac{2}{5}\right) \div \left(\frac{10}{25}\right)\right\}^{-3}$ c. $\frac{\{(3-2)^3 - 5^2\}^4}{6^0}$ d. $\frac{[\{(4+2)^3 \times 3^2\}^0 - 7^2]^2}{(5^2 - 3^2)^2}$
8. **Evaluate the following:**
   a. $(5^0 - 7^0) \times 3^3 \div 6^0$ b. $\left\{\frac{(5 \times 3)^0 \div 6^0}{7^0}\right\}^3$ c. $\left[\left\{\left(\frac{-66}{19}\right)^0\right\}^8\right]^{-7}$ d. $\frac{5^8 \times b^7}{(25)^3 \times b^4}$
9. **Express the numbers appearing in the following statements in the scientific notation.**
   a. There are 86,400 seconds in a day.
   b. Mass of the Moon is 73,49,00,00,00,00,00,00,00,00,000 kg.
   c. Distance between the Sun and the Earth is 1,46,90,00,00,000 m.
   d. Mass of the Jupiter is 1,89,87,00,00,00,00,00,00,00,00,00,00,000 kg.
10. **Express the following numbers in the expanded form.**
    a. 2.36×10^9 b. 5.6×10^{11} c. 0.8×10^{-5} d. 2.1875×10^{-6}

INTRODUCTION TO EXPONENTS

In the previous class, we have learnt that exponents are a shortened way to show how many times a number is multiplied by itself.

Example: a. $5 \times 5 \times 5 \times 5 \times 5 \times 5 = 5^6$ b. $(-6) \times (-6) \times (-6) \times (-6) \times (-6) = (-6)^5$

c. $\frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = \left(\frac{3}{4}\right)^4$ d. $\left(\frac{-7}{8}\right) \times \left(\frac{-7}{8}\right) \times \left(\frac{-7}{8}\right) \times \left(\frac{-7}{8}\right) \times \left(\frac{-7}{8}\right) = \left(\frac{-7}{8}\right)^5$

The representation or notation for expressing the product of an integer with itself several times is known as **exponential notation** or **power notation**.

For example, in $\left(\frac{-7}{8}\right) \times \left(\frac{-7}{8}\right) \times \left(\frac{-7}{8}\right) \times \left(\frac{-7}{8}\right) \times \left(\frac{-7}{8}\right) = \left(\frac{-7}{8}\right)^5$ ← Exponent, ← Base

It is read as $\frac{-7}{8}$ raised to the power 5.

In general, if a is any rational number, then $a \times a \times a \times a \times a \times \ldots \times a$ (n times) $= a^n$.

Here, a^n is called the nth power of a and is also read as **a raised to the power of n.**

The rational number a is called the **base** and n is called the **exponent** or **power** or **index.**

Look at the following examples:

a. $(-1)^2 = (-1) \times (-1) = 1$

b. $(-1)^3 = (-1) \times (-1) \times (-1) = -1$

c. $(-2)^6 = (-2) \times (-2) \times (-2) \times (-2) \times (-2) \times (-2) = 64$

d. $(-2)^7 = (-2) \times (-2) \times (-2) \times (-2) \times (-2) \times (-2) \times (-2) = -$

e. $\left(\frac{-1}{4}\right)^4 = \left(\frac{-1}{4}\right) \times \left(\frac{-1}{4}\right) \times \left(\frac{-1}{4}\right) \times \left(\frac{-1}{4}\right) = \frac{1}{256}$

f. $\left(\frac{-1}{3}\right)^3 = \left(\frac{-1}{3}\right) \times \left(\frac{-1}{3}\right) \times \left(\frac{-1}{3}\right) = \frac{-1}{27}$

Also, we have

g. $\left(\frac{3}{4}\right)^3 = \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = \frac{3 \times 3 \times 3}{4 \times 4 \times 4} = \frac{3^3}{4^3} = \frac{27}{64}$

h. $\left(\frac{-4}{5}\right)^5 = \left(\frac{-4}{5}\right) \times \left(\frac{-4}{5}\right) \times \left(\frac{-4}{5}\right) \times \left(\frac{-4}{5}\right) \times \left(\frac{-4}{5}\right) = \frac{(-4) \times (-4) \times (-4) \times (-4) \times (-4)}{(5) \times (5) \times (5) \times (5) \times (5)} = \frac{(-4)^5}{5^5} = \frac{-1024}{3125}$

Remember

- A negative rational number raised to an odd power is negative, i.e., $(-1)^{\text{odd natural number}} = -1$.
- A negative rational number raised to an even power is positive, i.e., $(-1)^{\text{even natural number}} = 1$.
- If $\frac{p}{q}$ is any rational number and n is an integer, then $\left(\frac{p}{q}\right)^n = \frac{p^n}{q^n}$. For example, $\left(\frac{5}{7}\right)^6 = \frac{5^6}{7^6} = \frac{15625}{117649}$.

NEGATIVE INTEGRAL EXPONENT

We know that $10^2 = 10 \times 10 = 100$; $10^1 = 10 = \frac{100}{10}$ and $10^0 = 1 = \frac{10}{10}$.

What will happen if the exponent becomes negative? $10^{-1} = ?$

In the above pattern, we observe that as the exponent decreases by 1, the value becomes one-tenth of the previ value. Continuing the above pattern, we get $10^{-1} = \frac{1}{10}$ and $10^{-2} = \frac{1}{10} \div 10 = \frac{1}{10} \times \frac{1}{10} = \frac{1}{10^2}$.

Let us follow a similar pattern for a base other than 10.

a. $(-3)^2 = 9$

$(-3)^1 = \frac{9}{-3} = -3$; $(-3)^0 = \frac{-3}{-3} = 1$; $(-3)^{-1} = \frac{1}{(-3)^1}$; $(-3)^{-2} = \frac{1}{(-3)^2}$

b. $\left(\frac{7}{9}\right)^2 = \frac{49}{81}$; $\left(\frac{7}{9}\right)^1 = \frac{\frac{49}{81}}{\frac{7}{9}} = \frac{49}{81} \times \frac{9}{7} = \frac{7}{9}$; $\left(\frac{7}{9}\right)^0 = \frac{\frac{7}{9}}{\frac{7}{9}} = \frac{7}{9} \times \frac{9}{7} = 1$; $\left(\frac{7}{9}\right)^{-1} = \frac{1}{\frac{7}{9}}$; $\left(\frac{7}{9}\right)^{-2} = \frac{1}{\left(\frac{7}{9}\right)^2}$

In general, if a is any non-zero rational number and n is an integer, then $a^{-n} = \frac{1}{a^n}$. We can say that a^{-n} is reciprocal of a^n. We can also say that if $\frac{a}{b}$ is any non-zero rational number and n is integer, then

Solved Examples

Example 1: Express each of the following as a rational number of the form, $\frac{p}{q}$:

a. 4^{-3} b. $\left(\frac{5}{3}\right)^{-3}$ c. $\frac{1}{2^{-5}}$

Solution: a. $4^{-3} = \frac{1}{4^3} = \frac{1}{64}$ $\left[\because a^{-n} = \frac{1}{a^n}\right]$

b. $\left(\frac{5}{3}\right)^{-3} = \frac{1}{\left(\frac{5}{3}\right)^3} = \left(\frac{3}{5}\right)^3 = \frac{3^3}{5^3} = \frac{27}{125}$ $\left[\because \left(\frac{p}{q}\right)^{-n} = \left(\frac{q}{p}\right)^n\right]$ $\left[\because \left(\frac{p}{q}\right)^n = \frac{p^n}{q^n}\right]$

c. $\frac{1}{2^{-5}} = \frac{1}{\frac{1}{2^5}} = \frac{2^5}{1} = 32$ $\left[\because a^{-n} = \frac{1}{a^n}\right]$

Example 2: Express each of the following as a rational number of the from, $\frac{p}{q}$:

a. $\left(\frac{5}{8}\right)^{-2} \times \left(\frac{4}{3}\right)^{-3}$ b. $\left(\frac{-2}{7}\right)^{-4} \times \left(\frac{-7}{9}\right)^3$

Solution : a. Here, $\left(\frac{5}{8}\right)^{-2} \times \left(\frac{4}{3}\right)^{-3} = \frac{1}{\left(\frac{5}{8}\right)^2} \times \frac{1}{\left(\frac{4}{3}\right)^3} = \frac{1}{\frac{5^2}{8^2}} \times \frac{1}{\frac{4^3}{3^3}} = \frac{8^2}{5^2} \times \frac{3^3}{4^3} = \frac{64}{25} \times \frac{27}{64}$ $\left[\because \left(\frac{p}{q}\right)^{-n} = \left(\frac{q}{p}\right)^n\right]$

$= \frac{64 \times 27}{25 \times 64} = \frac{27}{25}$

b. Here, $\left(\frac{-2}{7}\right)^{-4} \times \left(\frac{-7}{9}\right)^3 = \frac{1}{\left(\frac{-2}{7}\right)^4} \times \left(\frac{-7}{9}\right)^3 = \frac{1}{\frac{(-2)^4}{7^4}} \times \frac{(-7)^3}{9^3} = \frac{7^4}{2^4} \times \frac{-7^3}{9^3} = \frac{2401}{16} \times \frac{-343}{729}$ $\left[\because a^{-n} = \frac{1}{a^n}\right]$

$= \frac{2401 \times (-343)}{16 \times 729} = \frac{-823543}{11664}$

Example 3: Simplify:

a. $(4^{-1} + 8^{-1}) \div \left(\frac{2}{3}\right)^{-1}$ b. $(2^{-1} \div 5^{-1}) \div \left(\frac{-5}{8}\right)^{-1}$

Solution: a. Here, $(4^{-1} + 8^{-1}) \div \left(\frac{2}{3}\right)^{-1} = \left(\frac{1}{4} + \frac{1}{8}\right) \div \frac{1}{\frac{2}{3}}$

LCM of 4 and 8 is 8 $= \left(\frac{2+1}{8}\right) \times \frac{2}{3} = \frac{3}{8} \times \frac{2}{3} = \frac{3 \times 2}{8 \times 3} = \frac{1}{4}$

b. Here, $(2^{-1} \div 5^{-1})^2 \div \left(\frac{-5}{8}\right)^{-1} = \left(\frac{1}{2} \div \frac{1}{5}\right)^2 \div \frac{1}{\frac{-5}{8}}$ $\left[\because a^{-n} = \frac{1}{a^n}\right]$

$= \left(\frac{1}{2} \times \frac{5}{1}\right)^2 \times \frac{8}{-5} = \left(\frac{5}{2}\right)^2 \times \frac{8}{-5} = \frac{5^2}{2^2} \times \frac{8}{-5}$ $\left[\because \left(\frac{p}{q}\right)^n = \frac{p^n}{q^n}\right]$

$= \frac{25}{4} \times \frac{8}{-5} = \frac{25 \times (-8)}{4 \times 5} = -10$

Example 4: Simplify:

a. $\left(\frac{1}{4}\right)^{-2} + \left(\frac{1}{2}\right)^{-2} + \left(\frac{1}{3}\right)^{-2}$ b. $\left\{6^{-1} + \left(\frac{3}{2}\right)^{-1}\right\}^{-1}$

Solution: a. Here, $\left(\frac{1}{4}\right)^{-2} + \left(\frac{1}{2}\right)^{-2} + \left(\frac{1}{3}\right)^{-2} = \frac{1}{\left(\frac{1}{4}\right)^2} + \frac{1}{\left(\frac{1}{2}\right)^2} + \frac{1}{\left(\frac{1}{3}\right)^2} = \frac{1}{\frac{1^2}{4^2}} + \frac{1}{\frac{1^2}{2^2}} + \frac{1}{\frac{1^2}{3^2}}$ $\left[\because a^{-n} = \frac{1}{a^n}\right]$

$= \frac{4^2}{1^2} + \frac{2^2}{1^2} + \frac{3^2}{1^2} = \frac{16}{1} + \frac{4}{1} + \frac{9}{1} = 16 + 4 + 9 = 29$

b. Here, $\left\{6^{-1} + \left(\frac{3}{2}\right)^{-1}\right\}^{-1} = \left\{\frac{1}{6} + \frac{1}{\frac{3}{2}}\right\}^{-1} = \left\{\frac{1}{6} + \frac{2}{3}\right\}^{-1} = \left\{\frac{1+4}{6}\right\}^{-1} = \left(\frac{5}{6}\right)^{-1} = \frac{1}{\frac{5}{6}} = \frac{6}{5}$ $\left[\because a^{-n} = \frac{1}{a^n}\right]$

Example 5 : By what number should $(-6)^{-1}$ be multiplied so that the product may be equal to $(15)^{-1}$?

Solution : Let the number to be multiplied be a.

Now, $(-6)^{-1} \times a = (15)^{-1}$

$$\Rightarrow \frac{1}{-6} \times a = \frac{1}{15} \Rightarrow a = \frac{1}{15} \div \frac{1}{-6} \Rightarrow a = \frac{1}{15} \times \frac{-6}{1} = \frac{-6}{15} = \frac{-2}{5} \qquad \left[\because a^{-n} = \frac{1}{a^n}\right]$$

Thus, the required number is $\frac{-2}{5}$.

LAWS OF EXPONENTS WITH INTEGRAL POWERS

Rule 1 : If a is a rational number and m and n are two integers (positive or negative), then

$$a^m \times a^n = a^{m+n}$$

This is known as **product rule**.

Rule 2 : If a is a rational number and m and n are two integers (positive or negative), then

$$a^m \div a^n = a^{m-n}, \text{ where } m > n$$

This is known as **quotient rule**.

Rule 3 : If a is a non-zero rational number, and m and n are two integers (positive or negative), then

$$(a^m)^n = a^{mn}$$

This is called **power rule**.

Rule 4 : If a and b are two non-zero rational numbers and m is an integer (positive or negative), then

$$a^m \times b^m = (ab)^m$$

Rule 5 : If a is a non-zero rational number, then $a^0 = 1$.

Example 6: Simplify each of the following :

a. $(-2)^{-3} \times (-2)^{-4}$ b. $3^2 \times 3^{-5} \times 3^6$ c. $2^{-7} \div 2^{-3}$

d. $(4^2)^{-3}$ e. $3^{-3} \times (-7)^{-3}$ f. $\frac{3^{-5}}{4^{-5}}$

Solution : (a) We have $(-2)^{-3} \times (-2)^{-4} = (-2)^{-3+(-4)} = (-2)^{-7} = \frac{1}{(-2)^7} = \frac{1}{-128}$ $[\because a^m \times a^n = a^{m+n}]$

(b) We have $3^2 \times 3^{-5} \times 3^6 = 3^{2+(-5)+6} = 3^3 = 27$ $[\because a^m \times a^n \times a^p = a^{m+n+p}]$

(c) We have $2^{-7} \div 2^{-3} = 2^{-7-(-3)} = 2^{-7+3} = 2^{-4} = \frac{1}{2^4} = \frac{1}{16}$ $[\because a^m \div a^n = a^{m-n}]$

(d) We have $(4^2)^{-3} = 4^{2 \times (-3)} = 4^{-6} = \frac{1}{4^6} = \frac{1}{4096}$ $[\because (a^m) = a^{mn}]$

(e) We have $3^{-3} \times (-7)^{-3} = [3 \times (-7)]^{-3} = (-21)^{-3} = \frac{1}{(-21)^3} = \frac{1}{-9261}$ $[\because a^m \times b^m = (a \times b)^m]$

(f) We have $\frac{3^{-5}}{4^{-5}} = \left(\frac{3}{4}\right)^{-5} = \left(\frac{4}{3}\right)^5 = \frac{4^5}{3^5} = \frac{1024}{243}$ $\left[\because \frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m\right]$

Example 7: Solve:

a. $\left(\frac{-5}{6}\right)^2 \times \left(\frac{-5}{6}\right)^{-5} \times \left(\frac{-5}{6}\right)^4$ b. $\left(\frac{9}{14}\right)^{-4} \div \left(\frac{9}{14}\right)^{-6}$ c. $\left(\frac{6}{17}\right)^{-2} \div \left(\frac{6}{17}\right)^3$

d. $\left\{\left(\frac{11}{15}\right)^{-4}\right\}^{-3}$ e. $(-3)^4 \times \left(\frac{5}{3}\right)^4$ f. $(8)^{-3} \times 4^6 \times (2)^3$

Solution : a. We have $\left(\frac{-5}{6}\right)^2 \times \left(\frac{-5}{6}\right)^{-5} \times \left(\frac{-5}{6}\right)^4 = \left(\frac{-5}{6}\right)^{2+(-5)+4} = \left(\frac{-5}{6}\right)^1 = \frac{-5}{6}$ $[\because a^m \times a^n \times a^p = a^{m+n+p}]$

b. We have $\left(\frac{9}{14}\right)^{-4} \div \left(\frac{9}{14}\right)^{-6} = \left(\frac{9}{14}\right)^{-4-(-6)} = \left(\frac{9}{14}\right)^{-4+6} = \left(\frac{9}{14}\right)^2 = \frac{9^2}{14^2} = \frac{81}{196}$ $[\because a^m \div a^n = a^{m-n}]$ $\left[\because \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right]$

c. We have $\left(\frac{6}{17}\right)^{-2} \div \left(\frac{6}{17}\right)^{3} = \left(\frac{6}{17}\right)^{-2-3} = \left(\frac{6}{17}\right)^{-5}$ $[\because a^m \div a^n = a^{m-n}]$

$= \frac{1}{\left(\frac{6}{17}\right)^{5}} = \frac{1}{\frac{6^5}{17^5}} = \frac{17^5}{6^5} = \frac{1419,857}{7776}$ $\left[\because \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right]$

d. We have $\left\{\left(\frac{11}{15}\right)^{-4}\right\}^{-3} = \left(\frac{11}{15}\right)^{-4\times(-3)} = \left(\frac{11}{15}\right)^{12}$ $[\because (a^m)^n = a^{mn}]$

e. We have $(-3)^4 \times \left(\frac{5}{3}\right)^4 = \left(-3 \times \frac{5}{3}\right)^4 = (-5)^4 = 5^4 = 625$ $[\because a^m \times b^n = (a\times b)^m]$

f. We have $(8)^{-3} \times 4^0 \times (2)^3 = \frac{1}{8^3} \times 1 \times 2^3 = \frac{8}{512} = \frac{1}{64}$ $\left[\because a^{-m} = \frac{1}{a^m} \text{ and } a^0 = 1\right]$

Example 8: Solve:

a. $\left\{\left(\frac{1}{3}\right)^{-2} - \left(\frac{1}{2}\right)^{-3}\right\} \div \left(\frac{1}{4}\right)^{-2}$ b. $\left\{\left(\frac{6}{7}\right)^{2}\right\}^{3} \times \left\{\left(\frac{6}{7}\right)^{-4}\right\}^{2}$ c. $\left\{\left(\frac{3}{7}\right)^{4} \times \left(\frac{3}{7}\right)^{3}\right\} \div \left(\frac{27}{343}\right)^{3}$

Solution: a. We have $\left\{\left(\frac{1}{3}\right)^{-2} \times \left(\frac{1}{2}\right)^{-3}\right\} \div \left(\frac{1}{4}\right)^{-2} = \left\{\frac{1}{\left(\frac{1}{3}\right)^{2}} - \frac{1}{\left(\frac{1}{2}\right)^{3}}\right\} \div \frac{1}{\left(\frac{1}{4}\right)^{2}}$ $\left[\because a^{-m} = \frac{1}{a^m}\right]$

$= \left\{\frac{1}{\frac{1^2}{3^2}} - \frac{1}{\frac{1^3}{2^3}}\right\} \div \frac{1}{\frac{1^2}{4^2}}$ $\left[\because \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}\right]$

$= \left\{\frac{3^2}{1^2} - \frac{2^3}{1^3}\right\} \div \frac{4^2}{1^2} = \{9 - 8\} \div 16 = 1 \div 16 = \frac{1}{16}$

b. We have $\left\{\left(\frac{6}{7}\right)^{2}\right\}^{3} \times \left\{\left(\frac{6}{7}\right)^{-4}\right\}^{2} = \left(\frac{6}{7}\right)^{2\times3} \times \left(\frac{6}{7}\right)^{-4\times2}$ $[\because (a^m)^n = a^{mn}]$

$= \left(\frac{6}{7}\right)^{6} \times \left(\frac{6}{7}\right)^{-8} = \left(\frac{6}{7}\right)^{6+(-8)}$ $[\because a^m \times a^n = a^{m+n}]$

$= \left(\frac{6}{7}\right)^{-2} = \frac{1}{\left(\frac{6}{7}\right)^{2}} = \frac{1}{\frac{6^2}{7^2}} = \frac{7^2}{6^2} = \frac{49}{36}$

c. We have $\left\{\left(\frac{3}{7}\right)^{4} \times \left(\frac{3}{7}\right)^{3}\right\} \div \left(\frac{27}{343}\right)^{3} = \left(\frac{3}{7}\right)^{4+3} \div \left(\frac{3^3}{7^3}\right)^{3} = \left(\frac{3}{7}\right)^{7} \div \frac{3^{3\times3}}{7^{3\times3}} = \left(\frac{3}{7}\right)^{7} \div \frac{3^9}{7^9}$

$= \left(\frac{3}{7}\right)^{7} \div \left(\frac{3}{7}\right)^{9} = \left(\frac{3}{7}\right)^{7-9} = \left(\frac{3}{7}\right)^{-2} = \frac{1}{\left(\frac{3}{7}\right)^{2}} = \frac{1}{\frac{3^2}{7^4}} = \left(\frac{7^2}{3^2}\right) = \frac{49}{9}$

Example 9: Evaluate:

a. $\frac{8^{-1}\times5^3}{2^{-4}}$ b. $\frac{3^{-5}\times10^{-5}\times125}{5^{-7}\times6^{-5}}$ c. $\frac{a^{-2}\times b^{-3}\times c^{-4}\times d^2}{a^2\times b^{-4}\times c^{-2}\times d^2}$

Solution: a. We have $\frac{8^{-1}\times5^3}{2^{-4}} = \frac{(2^3)^{-1}\times5^3}{2^{-4}} = \frac{2^{3\times-1}\times5^3}{2^{-4}} = \frac{2^{-3}\times5^3}{2^{-4}}$

$= 2^{-3-(-4)} \times 5^3 = 2^{-3+4} \times 5^3 = 2^1 \times 5^3 = 2 \times 125 = 250$

b. We have $\frac{3^{-5}\times10^{-5}\times125}{5^{-7}\times6^{-5}} = \frac{3^{-5}\times(2\times5)^{-5}\times(5)^3}{5^{-7}\times(2\times3)^{-5}} = \frac{3^{-5}\times2^{-5}\times5^{-5}\times5^3}{5^{-7}\times2^{-5}\times3^{-5}}$

$= 3^{-5-(-5)} \times 2^{-5-(-5)} \times 5^{-5+3-(-7)} = 3^{-5+5} \times 2^{-5+5} \times 5^{-5+3+7}$

$= 3^0 \times 2^0 \times 5^5 = 1 \times 1 \times 3,125 = 3,125$

c. We have $\frac{a^{-2}\times b^{-3}\times c^{-4}\times d^2}{a^2\times b^{-4}\times c^{-2}\times d^2} = a^{-2-2} \times b^{-3-(-4)} \times c^{-4-(-2)} \times d^{2-2} = a^{-4} \times b^{-3+4} \times c^{-4+2} \times d^0$

$= a^{-4} \times b^1 \times c^{-2} \times 1 = \frac{b}{a^4 \times c^2} = \frac{b}{a^4c^2}$

Example 10: By what number should $\left(\frac{-3}{2}\right)^{-3}$ be divided so that the quotient may be $\left(\frac{4}{27}\right)^{-2}$?

Solution : Let the required number be x. Then, $\left(\frac{-3}{2}\right)^{-3} \div x = \left(\frac{4}{27}\right)^{-2} \Rightarrow \frac{1}{\left(\frac{-3}{2}\right)^{3}} \times \frac{1}{x} = \frac{1}{\left(\frac{4}{27}\right)^{2}} \Rightarrow \frac{1}{\left(\frac{-3}{2}\right)^{3}} \div \frac{1}{\left(\frac{4}{27}\right)^{2}} = x$

$$\Rightarrow x = \frac{\frac{1}{(-3)^3}}{\frac{1}{2^3}} \div \frac{\frac{1}{4^2}}{\frac{1}{27^2}} = \frac{2^3}{(-3)^3} \div \frac{27^2}{4^2} = \frac{8}{-27} \div \frac{729}{16} = \frac{8}{-27} \times \frac{16}{729} = \frac{-8 \times 16}{27 \times 729} = \frac{-128}{19683}$$

Thus, the required number is $\frac{-128}{19683}$.

Example11: Find the value of x in each of the following:

a. $\left(\frac{6}{7}\right)^{x} \times \left(\frac{6}{7}\right)^{5} = \left(\frac{6}{7}\right)^{5}$ b. $\left(\frac{1}{2}\right)^{2x} \div \left(\frac{1}{2}\right)^{2} = \left(\frac{1}{2}\right)^{4}$

Solution : a. We have $\left(\frac{6}{7}\right)^{x} \times \left(\frac{6}{7}\right)^{5} = \left(\frac{6}{7}\right)^{5} \Rightarrow \left(\frac{6}{7}\right)^{x+5} = \left(\frac{6}{7}\right)^{5}$

Comparing the powers of both sides, as the bases are same, we have $x + 5 = 5 \Rightarrow x = 5 - 5 = 0$.

b. We have, $\left(\frac{1}{2}\right)^{2x} \div \left(\frac{1}{2}\right)^{2} = \left(\frac{1}{2}\right)^{4} \Rightarrow \left(\frac{1}{2}\right)^{2x-2} = \left(\frac{1}{2}\right)^{4}$

Comparing the powers of both sides, as the bases are same, we have

$2x - 2 = 4 \Rightarrow 2x = 2 + 4 \Rightarrow 2x = 6 \Rightarrow x = 3$

EXERCISE 2(A)

1. Express each of the following as a rational number of the form $\frac{p}{q}$:

a. $\left(\frac{3}{7}\right)^{2}$ b. $\left(\frac{3}{4}\right)^{5}$ c. $\left(\frac{-2}{3}\right)^{4}$ d. $\left(\frac{-5}{9}\right)^{3}$

2. Simplify the following:

a. $\left(\frac{4}{5}\right)^{2} \times \left(\frac{-2}{3}\right)^{3}$ b. $\left(\frac{-1}{3}\right)^{5} \div \left(\frac{2}{3}\right)^{3}$ c. $(-2)^{5} \div \left(\frac{-1}{3}\right)^{3}$ d. $\left(\frac{7}{3}\right)^{3} \times \left(\frac{7}{3}\right)^{-5}$

3. Express each of the following as a rational number in the form $\frac{p}{q}$:

a. 5^{-3} b. $(-4)^{-2}$ c. $\frac{1}{7^{-2}}$ d. $\left(\frac{1}{3}\right)^{-5}$

e. $\left(\frac{2}{3}\right)^{-2}$ f. $\left(\frac{5}{8}\right)^{3}$ g. $(-7)^{-4}$ h. $\left(\frac{-3}{2}\right)^{-2}$

4. Find the values:

a. $4^{-1} + 5^{-1}$ b. $(2^{-1} + 9^{-1}) \times 3^{2}$ c. $(2^{-1} \times 4^{-1}) \div 2^{-2}$ d. $(5^{-1} \times 2^{-1}) \div 6^{-1}$

5. Simplify:

a. $\left(\frac{1}{2}\right)^{-1} + \left(\frac{1}{3}\right)^{-1} + \left(\frac{1}{4}\right)^{-1}$ b. $\left(\frac{1}{2}\right)^{-2} + \left(\frac{1}{3}\right)^{-2} + \left(\frac{1}{4}\right)^{-2}$ c. $\left[\left(\frac{1}{3}\right)^{-3} - \left(\frac{1}{2}\right)^{-3}\right] \div \left(\frac{1}{4}\right)^{-3}$ d. $(2^{2} + 3^{2} - 4^{2}) \div \left(\frac{3}{2}\right)^{2}$

6. Simplify:

a. $\left(\frac{2}{3}\right)^{3} \times \left(\frac{2}{3}\right)^{4}$ b. $\left(\frac{-4}{5}\right)^{9} \div \left(\frac{-4}{5}\right)^{7}$ c. $(-4)^{6} \div (-4)^{8}$ d. $\left(\frac{2}{3}\right)^{3} \times \left(\frac{2}{3}\right)^{-5}$

e. $\left(\frac{1}{2^{3}}\right)^{2}$ f. $\left[\left(\frac{2}{5}\right)^{-3}\right]^{2}$ g. $\left(\frac{-2}{3}\right)^{-4} \times \left(\frac{1}{8}\right)^{-4}$ h. $\left(\frac{-4}{7}\right)^{7} \times \left(\frac{-4}{5}\right)^{-7} \times \left(\frac{-4}{5}\right)^{-7}$

7. Fill in the blanks:

a. $2^{3} \times 2^{4} = 2$ b. $(-4)^{5} \times (-4)^{6} = (-4)$ c. $\left(\frac{2}{3}\right)^{7} \times \left(\frac{2}{3}\right)^{4} = \left(\frac{2}{3}\right)$ d. $\left(\frac{3}{4}\right)^{x} \div \left(\frac{3}{4}\right)^{5} = \left(\frac{3}{4}\right)$

e. $(-4)^{7} \div (-4)^{3} = (-4)$ f. $\left(\frac{-3}{7}\right)^{7} \div \left(\frac{-3}{7}\right)^{3} = \left(\frac{-3}{7}\right)$ g. $8^{13} \div 8^{19} = \frac{1}{8}$ h. $(-4)^{11} \div (-4)^{15} = \frac{1}{(-4)}$